Data Structures ***Python Assignment 1***

Spring 2026

Objective

The goal of this assignment is to gain initial experience with the Python programming language, work with code, and run experiments to compare different sorting algorithms in terms of running time.

Submission Instructions (Moodle)

Submit a link to a GitHub repository containing your code.

GitHub Repository Structure

Create a repository with the following structure:

Repository name: `Sorting_Assignment`

Inside the repository:

- run_experiments.py – main code file
- README.md – short explanation file
- result1.png and result2.png – the two required plots
  (You may add additional plots in a subfolder.)

Assignment Requirements

Part A – Algorithm Implementation

Choose three sorting algorithms from the following list:

1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort

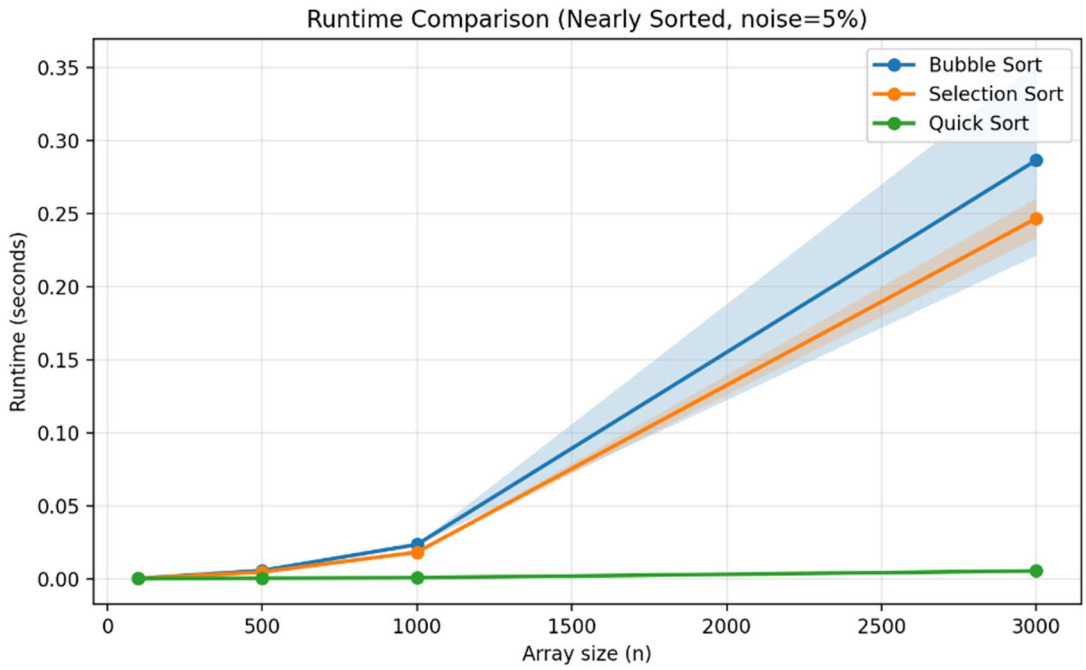
Part B – Comparative Experiment (Random Arrays)

Your task is to compare the running times of the three algorithms you selected.

For the experiment:

1. Generate arrays of different sizes, each containing random integers. Example sizes:
    - 100, 500, 1,000, 5,000, ...
    - up to 1,000,000 elements
2. Measure the running time of each algorithm.
3. Repeat each experiment multiple times for every array size.
4. Report:
    - Average running time
    - Standard deviation of the running time
5. Present the results in a comparison plot named result1.png.

The goal is to illustrate the difference in growth rate between slower and faster algorithms, similar to theoretical complexity plots.



Part C – Experiment with Noise or Partial Order (Nearly Sorted Arrays)

Perform an additional experiment where:

1. You start with a sorted array.
2. You add noise (partial shuffling of the elements).
3. You compare the running times again and save the plot as result2.png.

Example noise levels:

- Swap 5% of the elements randomly
- Swap 20% of the elements randomly

Part D – Simple User Interface (Command Line Arguments)

Your program must support running experiments from the command line.

Example command:

```
python run_experiments.py -a 1 2 5 -s 100 500 3000 -e 1 -r 20
```

Your interface must allow the user to choose:

- Which algorithms to compare (-a)
- Array sizes (-s)
- Experiment type / noise level (-e)
    - 1 – Nearly sorted with 5% noise
    - 2 – Nearly sorted with 20% noise
- Number of repetitions (-r)

Algorithm IDs:

- 1 – Bubble Sort
- 2 – Selection Sort
- 3 – Insertion Sort
- 4 – Merge Sort
- 5 – Quick Sort

README File

In README.md, include:

- Student name(s)
- The selected algorithms
- Present the two figures (result1.png and result2.png) and below each figure add a short explanation of the results.
- For result2.png, explain if and how the running times changed and why.

How to Create a GitHub Repository (for Beginners)

1. Go to GitHub.
2. Log in or create an account.
3. Click New Repository.
4. Name it: Sorting_Assignment
5. Select:
    - Public
    - Add README
6. Click Create Repository.
7. Upload your files to the repository.